

A Large-Scale Constrained Multi-Objective Evolutionary Algorithm with Promising Region Detection and Diversity Enhancement

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Abstract—Most existing constrained multi-objective evolutionary algorithms (CMOEAs) suffer from slow convergence and may even fail to find feasible solutions when dealing with problems that involve large-scale decision spaces and complex constraints. To this end, this paper introduces a promising region-guided CMOEA. In the proposed algorithm, a promising region detection and diversity enhancement strategy is designed to avoid entrapment in local optima. Specifically, this strategy first utilizes well-converged feasible non-dominated solutions among all solutions examined so far to detect the promising region where the constrained Pareto front may exist. Then, three non-dominated sorting procedures based on extended Pareto dominance, extended reverse Pareto dominance, and classical Pareto dominance, are sequentially executed to guide the population to evenly search the promising region and approach the constrained Pareto front from diverse search directions. In addition, an accelerated evolutionary search strategy is devised to improve reproduction quality in large-scale search spaces. Comprehensive experiments conducted on four benchmark test suites and thirty coal mine integrated energy system dispatch problems demonstrate the superiority of the proposed algorithm over seven state-of-the-art algorithms.

Index Terms—Constrained multi-objective optimization, large-scale optimization, promising region, accelerated evolutionary search.

I. INTRODUCTION

MANY engineering applications require the simultaneous optimization of multiple conflicting objective functions while ensuring all constraints are satisfied. For example, cascade reservoir scheduling [1], job shop scheduling [2], and the dispatch optimization of coal mine integrated energy systems (CMIES) [3], [4]. These problems are collectively known as constrained multi-objective optimization problems (CMOPs) [5], and those with 100 or more decision variables are usually termed large-scale CMOPs (LSCMOPs) [6], [7].

To cope with constraints, numerous constraint handling techniques (CHTs) have been proposed [5], including the

constraint dominance principle (CDP) [8], ε -constrained methods [9], penalty function-based methods [10], and multi-objective-based methods [11]. These CHTs are embedded into multi-objective evolutionary algorithms (MOEAs) to form various constrained MOEAs (CMOEAs). Recently, algorithm designers have drawn inspiration from the divide-and-conquer idea and developed multi-stage-based CMOEAs [12]–[14] and multi-population-based CMOEAs [15]–[18]. In these methods, different stages or populations employ distinct CHTs and emphasize varying priorities such as convergence, diversity, and feasibility. However, most CMOEAs neglect the scalability of decision variables and the adopted CHTs lack effective diversity enhancement capabilities. When applied to LSCMOPs involving hundreds of variables, their effectiveness could potentially deteriorate sharply [19], as the properties of some LSCMOPs (e.g., variable interactions, constraint boundaries, and fitness landscapes) may become more complex due to the “curse of dimensionality” [4], [6].

Moreover, most CMOEAs utilize conventional evolutionary operators (e.g., differential evolution (DE) [20], simulated binary crossover (SBX) [21], and polynomial mutation (PM) [22]) to explore the original variable space, which often results in low-quality offspring solutions and slow convergence when solving LSCMOPs [19]. Indeed, delicate search strategies have been tailored to strengthen the search capability, including variable grouping-based cooperative coevolution [23], problem transformation [24], decision space reduction [25], and enhanced reproduction mechanisms [26], [27]. However, these techniques rarely take the challenges posed by constraints into account. For example, the search space of the reformulated problem or the subspace formed by certain variables may not contain any feasible regions, as explained in Section S-I of the supplementary file.

Given the prevalence of LSCMOPs in many real-world applications, such as CMIES dispatch optimization [28] and optimal wireless sensor placement [29], it is imperative to develop effective large-scale CMOEAs (LSCMOEAs) that can mitigate the simultaneous impact of constraints and large-scale variables. To this end, this paper attempts to mine the hidden information from the feasible non-dominated solutions (FNDSs), and proposes a new LSCMOEA, termed PRCEA, for solving LSCMOPs. In this paper, the FNDSs are defined as the non-dominated solutions among all feasible solutions examined so far during the evolution. The main contributions are threefold:

- 1) A promising region detection and diversity enhancement

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t strategy, termed PRDD, is proposed to improve the capability of capturing the complete constrained Pareto front (CPF). Specifically, PRDD first extensively explores the search space to seek a set of well-converged FNDSSs in close proximity to the CPF, which are then utilized to identify the promising region where CPF may exist. Afterward, three dominance relations: 1) extended Pareto dominance (EPD), 2) extended reverse Pareto dominance (ERPD), and 3) Pareto dominance (PD), are sequentially applied to select solutions for the next generation. Notably, EPD promotes a well-distributed population within the promising region, while ERPD and PD provide driving forces toward the promising region from various search directions.

- 2) An accelerated evolutionary search strategy, named AES, is designed by integrating inter-learning and intra-learning mechanisms¹. In particular, AES divides the parent population into a loser group and a winner group. The inter-learning mechanism guides each loser to learn from a winner, thereby accelerating convergence toward feasible or promising infeasible regions. Meanwhile, the intra-learning mechanism enables each winner to construct promising search directions in the decision space and generate offspring along these directions to escape local optima and discover new feasible regions.
- 3) Based on the above two innovations, a promising region-guided LSCMOEA, called PRCEA, is proposed. Systematic experiments on four benchmark suites and thirty CMIES dispatch optimization problems validate the superior performance of PRCEA over seven state-of-the-art algorithms.

The remainder of this paper is structured as follows. Section II presents a concise literature review and outlines the research motivation of this study. Section III details the proposed PRCEA. The experimental setup is provided in Section IV, followed by the performance analysis of PRCEA and comparative studies in Section V and Section VI, respectively. Section VII concludes this paper and discusses potential avenues for future research.

II. RELATED WORK AND MOTIVATION

A. Basic Concepts

In general, the minimization form of a continuous CMOP can be expressed as:

$$\begin{aligned} \min \mathbf{f}(\mathbf{x}) &= (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_m(\mathbf{x}))^T, \mathbf{x} \in \Omega = \prod_{k=1}^n [l_k, u_k], \\ \text{s.t. } &\begin{cases} g_i(\mathbf{x}) \leq 0, & i = 1, \dots, p, \\ h_j(\mathbf{x}) = 0, & j = 1, \dots, q, \end{cases} \end{aligned} \quad (1)$$

where $\mathbf{x} = (x_1, \dots, x_n)^T$ is an n -dimensional solution in the decision (or search) space Ω , which is bounded by the lower bound $\mathbf{l} = (l_1, \dots, l_n)^T$ and upper bound $\mathbf{u} = (u_1, \dots, u_n)^T$.

¹In this paper, inter-learning refers to acquiring search guidance from other elite solutions, while intra-learning denotes constructing search directions from a solution's own information.

$\mathbf{f}(\mathbf{x})$ is an objective vector comprising m ($m \geq 2$) conflicting real-valued objective functions. $g_i(\mathbf{x})$ and $h_j(\mathbf{x})$ denote the i^{th} inequality and j^{th} equality constraints, respectively, with a total of p inequality constraints and q equality constraints. In particular, this paper considers the problem (1) with $n \geq 100$ and $m \in \{2, 3\}$, which is generally called an LSCMOP [7], [19]. For a solution $\mathbf{x} \in \Omega$, its overall constraint violation is computed as:

$$\phi(\mathbf{x}) = \sum_{i=1}^p \max\{0, g_i(\mathbf{x})\} + \sum_{j=1}^q \max\{0, |h_j(\mathbf{x})| - \delta\}, \quad (2)$$

where $\delta > 0$ is a tolerance parameter (e.g., 10^{-6}) that allows for small deviations from strict equality. $\phi(\mathbf{x}) = 0$ indicates that $\mathbf{x}$ is feasible; otherwise, $\mathbf{x}$ is infeasible.

For two decision vectors $\mathbf{x}, \mathbf{y} \in \Omega$, $\mathbf{x}$ is said to Pareto dominate $\mathbf{y}$ (denoted as $\mathbf{x} \prec \mathbf{y}$) if $\mathbf{f}(\mathbf{x}) \neq \mathbf{f}(\mathbf{y})$ and $f_k(\mathbf{x}) \leq f_k(\mathbf{y})$ for all $k \in \{1, \dots, m\}$. A solution $\mathbf{x}^* \in \Omega$ is unconstrained Pareto optimal if $\nexists \mathbf{x} \in \Omega$ such that $\mathbf{x} \prec \mathbf{x}^*$. In the objective space, the image mapped by all unconstrained Pareto optimal solutions is called the unconstrained Pareto front (UPF).

In addition, according to CDP [8], $\mathbf{x}$ is said to constrained dominate $\mathbf{y}$ (denoted as $\mathbf{x} \prec_{\text{CDP}} \mathbf{y}$) if $\phi(\mathbf{x}) < \phi(\mathbf{y})$, or $\phi(\mathbf{x}) = \phi(\mathbf{y})$ and $\mathbf{x} \prec \mathbf{y}$. A solution $\mathbf{x}^* \in \Omega$ is constrained Pareto optimal if $\nexists \mathbf{x} \in \Omega$ such that $\mathbf{x} \prec_{\text{CDP}} \mathbf{x}^*$. The set of all constrained Pareto optimal solutions is called the constrained Pareto set (CPS), and its image in the objective space forms the CPF.

B. Existing CMOEAs

Existing CMOEAs mainly fall into the following four categories based on the characteristics of their CHTs:

The first category of CMOEAs adopts feasibility-oriented CHTs. A notable example is CDP, which builds upon Pareto dominance (PD) by prioritizing solutions with lower ϕ values. Owing to its simplicity and ease of implementation, CDP has been widely integrated into various PD-based MOEAs, such as NSGA-II [8] and SPEA2 [30]. The feasibility-first principle has also been extended to decomposition-based MOEAs (e.g., NSGA-III [31] and MOEA/D [32]) by prioritizing constraint satisfaction over scalarizing function minimization during sub-problem updates. However, a strong reliance on feasibility may cause the population to persist in deceptive feasible regions separated from the CPF, or stagnate in easily discoverable parts of the global optimal feasible regions. To this end, researchers have increasingly explored the use of promising infeasible solutions to help search for new feasible regions. Following efforts include localized CDP [33], which applies CDP only within the local niche to maintain the competitiveness of some infeasible solutions. Beyond selection-based CHTs, some researchers have investigated repair mechanisms. For instance, [34] presents a gradient-based repair operator to move infeasible solutions toward feasible regions, where the repair step is computed using the gradient of constraint violation. Notably, this repair operation is invoked with a probability less than one to balance computational cost and search diversity.

The second category of CMOEAs uses problem transformation-based CHTs. For example, multi-objective-based methods [11] typically extend the overall constraint violation as a new objective, thereby formulating the following unconstrained multi-objective optimization problem (MOP):

$$\min \mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_m(\mathbf{x}), \phi(\mathbf{x}))^T, \mathbf{x} \in \Omega. \quad (3)$$

The Pareto optimal solutions of Eq. (3) are uniformly distributed between the CPF and UPF, thus solving this problem can effectively enhance global exploration and avoid getting trapped in local optima. The overall constraint violation can also be customized as various forms of penalty terms. ShiP [10] introduces a shift-based penalty function to shift infeasible solutions within the niche formed by their neighboring feasible solutions. In [35], a Kriging-based constrained probabilistic Pareto dominance is proposed, which redefines each objective of a solution as the negative product of its probability of feasibility and its probability of outperforming the compared solution. In addition, a series of simplified CMOPs can be created by pruning partial constraints from Eq. (1), among which the most prominent is the constraint-ignored MOP [15]:

$$\min \mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_m(\mathbf{x}))^T, \mathbf{x} \in \Omega. \quad (4)$$

ε methods [9] adopt a dynamic threshold ε to relax the constraint boundary, thereby formulating the following dynamic CMOP:

$$\begin{aligned} \min \mathbf{f}(\mathbf{x}) &= (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_m(\mathbf{x}))^T, \mathbf{x} \in \Omega, \\ \text{s.t. } \phi(\mathbf{x}) &\leq \varepsilon \end{aligned} \quad (5)$$

To promote local exploitation around the CPF, a novel dynamic CMOP is proposed in [20] by hybridizing the strengths of problems (3) and (5).

The methods in the third category divide the evolutionary process into multiple stages to alternatively switch optimization preferences. A well-known work is the push and pull search (PPS) pattern [36], in which the population is first pushed toward the UPF without constraints, and then pulled back to the CPF by shifting the focus to constraint satisfaction. Solvers developed following this pattern include CEMTFB [12] and DPTPEA [37]. Researchers have extended the PPS pattern through constraint pruning, as in C3M [13]. Specifically, during the pull phase, constraints are incrementally added to the optimization task until all are incorporated. Recent studies have explored partitioning the evolutionary process into multiple stages based on the population status (e.g., search stagnation or population distribution), which enables adaptive alternation of preferred optimization tasks upon entering a new stage. For example, CMOEA-MS [38] switches its search focus based on the feasibility ratio. LCMEA [39] utilizes a deep reinforcement learning (DRL) model to intelligently recommend the CHT at each generation, while CMODRL [40] employs the Actor-Critic framework to automatically configure the constraint relaxation degree.

The fourth category of CMOEAs collaboratively evolves multiple specialized populations to address the target CMOP and complementary helper problems, which are typically created using methods such as constraint pruning, constraint relax-

ation, and constraint transformation. Among constraint-pruned problems, the most widely used one is the constraint-ignored MOP 4, as adopted in EMC MO [15]. In these algorithms, the main task and helper task respectively search for the CPF and UPF, and enhance their problem-solving capabilities through knowledge transfer. Constraint-relaxed problems are considered in ILCMO [7] and IMTCMO [20], where the constraint boundary is relaxed to a dynamic ε -level. Constraint transformation methods reformulate constraints as penalty terms or as additional objective functions. For example, cDPEA [41] designs a self-adaptive penalty function to repair the objective vectors of infeasible solutions without altering their associated subregions. In TPEA [11], the innermost and outermost non-dominated solutions of Eq. (4) are used to provide escape and expansion forces, respectively. In addition, the combined use of multiple types of helper problems has been shown to significantly enhance algorithmic performance in solving various CMOPs. For instance, CMOC SO [42] customizes both constraint-ignored and constraint-relaxed problems to enhance global and local diversity, respectively. Recently, a growing body of research has integrated multi-stage designs into co-evolutionary frameworks. MCCMO [17] and MTOTC [18] introduce helper problems defined by different constraint subsets, and partially merge these subsets upon entering the next stage. Moreover, DRL-based optimization task recommendation systems have garnered considerable attention and have been successfully applied in CMOQLMT [43] to adaptively select the helper problem at each evolutionary stage.

The aforementioned CMOEAs primarily focus on designing effective CHTs and specialized mechanisms to address the scalability of constraint complexity, while few of them delve into the perplexity arising from large-scale variables. Consequently, the effectiveness of these CMOEAs may severely deteriorate when solving LSCMOPs due to the ‘‘curse of dimensionality’’.

C. Existing LSCMOEAs

The development of LSCMOEAs is still in its infancy, with most current efforts dedicated to designing enhanced reproduction mechanisms. CMOC SO [42] and POCEA [44] respectively introduce a competitive and cooperative swarm optimizer (CCSO) and a localized paired competitive swarm optimizer (CSO) to accelerate convergence while maintaining diversity. In IMTCMO-BS [6], a bidirectional sampling strategy is implemented using convergence-directed sampling and diversity-directed sampling. Notably, all three algorithms adopt ε methods to improve feasibility. LCMOE A [19] proposes a learnable DE guided by two improved vectors, which are derived from two well-trained multilayer perceptron models respectively for feasibility-first task and constraint-ignored task. The fitness of a solution is reflected by the weighted sum of its objective ranking and feasibility ranking. ILCMO [7] designs a dynamic indicator based on two relaxed constraint boundaries to promote uniform exploration of the potential CPF regions, and enhances reproduction via group-based intra- and inter-learning DE operators.

However, most of the above methods neglect the potential difficulty of finding feasible solutions in LSCMOPs, as scaling

up variables may lead to more intricate variable coupling relations and increasingly disconnected and narrow feasible regions [6]. Consequently, the adopted CHTs not only struggle with parameter tuning (e.g., the level of constraint relaxation and weights for objective and constraint rankings), but also lack the expansion force needed to spread solutions from one part of the global optimal feasible regions to other parts. Additionally, sampling directions may fail to intersect with the CPS, and the dimensionality-reduced search space might not contain any feasible solutions. Therefore, developing efficient LSCMOEAs remains a critical area for further research.

D. Motivation

Most existing CHTs can leverage infeasible solutions with good objective values to guide the population out of local optimal feasible regions and into easily accessible parts of the global optimal feasible regions. However, few CHTs are capable of providing sufficient expansion force to spread solutions to other narrow global optimal feasible regions. To intuitively understand this limitation, Figs. 1(a) to 1(d) show the effects of four widely used CHTs, respectively.

- Fig. 1(a) shows the impact of CDP, which unconditionally favors feasible solutions over infeasible ones, thereby driving the population quickly into feasible regions. However, if any feasible region remains uncovered, such as the narrow disconnected feasible region in the middle, it becomes nearly impossible to detect it, as solutions can hardly escape the detected feasible regions.
- As shown in Fig. 1(b), under the ε -level relaxation, the population leaves the feasible regions and approaches the ε -constrained relaxation boundary (dashed line). However, solutions struggle to spread into the middle feasible region since the detected relaxed feasible regions may not be large enough to connect with other undiscovered relaxed feasible regions.
- In Fig. 1(c), the population is directed to the UPF by ignoring constraints. However, the CPF may be separated from the UPF, thus it is impractical to discover all CPF segments by searching the UPF alone.
- Fig. 1(d) illustrates the multi-objective-based method for preserving non-dominated solutions of MOP (4), where the function ϕ is extended as an additional objective function. Although the maintained solutions are widely distributed in the objective space with good diversity, only a small portion of them approach the CPF. Therefore, this approach can only provide a limited expansion force to diffuse solutions into other undetected global optimal feasible regions (i.e., the middle feasible region).

In summary, the complete CPF is less likely to be obtained based on most existing CHTs.

For a solution $x \in \Omega$, the objective space can be partitioned into three disjoint regions:

- *Dominated Region*: $\{f(y) \in \mathbb{R}^m \mid x \prec y\}$;
- *Dominating Region*: $\{f(y) \in \mathbb{R}^m \mid y \prec x\}$;
- *Mutually Non-dominated Region*: $\{f(y) \in \mathbb{R}^m \mid x \not\prec y \wedge y \not\prec x\}$.

If x is a solution from the CPS, then the CPF must lie entirely within the mutually non-dominated region of x . Otherwise, there would exist another solution y from CPS such that either $x \prec y$ or $y \prec x$, which contradicts the mutual non-dominance property of constrained Pareto optimal solutions. Therefore, this paper defines the mutually non-dominated region induced by all identified constrained Pareto optimal solutions as the promising region, yet this region has received little attention in existing work. If an algorithm can extensively explore the search space to find the global optimal feasible regions and then focus on evenly searching the promising region, the entire CPF can be efficiently obtained regardless of the distribution of feasible regions and the relationship between the UPF and CPF. Furthermore, as the number of constrained Pareto optimal solutions increases or their diversity improves, the promising region shrinks, making the even search more precise.

Based on the above analysis, this paper proposes a new CHT, termed PRDD. Without prior knowledge of the problem, PRDD first extensively explores the entire search space to obtain a set of well-converged FNDSSs to locate the global optimal feasible regions. Subsequently, it focuses the search effort on the promising region to ensure a well-distributed population along the CPF. Fig. 1(e) illustrates the core idea of PRDD, where the promising region (highlighted in yellow) is defined as the mutually non-dominated region III of two FNDSSs (x_1 and x_2), whose objective vectors lie on the CPF. In PRDD, the quality of solutions in the promising region is evaluated by EPD to enhance diversity. Besides, in the dominating region I, solutions closer to the promising region or with lower ϕ values are preferred according to ERPD. While in the dominated region II, PD serves as the main selection criterion to emphasize convergence. Consequently, PRDD can effectively drive the population into the promising region from diverse search directions and ensure it is well-distributed, making it easier to detect the middle feasible region. Notably, while the promising region defined in both this paper and the literature [45], [46] contain region III, the definitions in [45], [46] differ by also including region I. As analyzed in Fig. 1(d), when the CPF is far from the UPF, wasting computational resources on a uniform search in region I provides only a limited expansion force for diffusing solutions to the middle feasible region.

Furthermore, this paper designs an AES reproduction approach to address the curse of dimensionality. Section III elaborates on the mechanisms of PRDD and AES, along with the newly developed PRCEA.

III. PROPOSED ALGORITHM

A. Outline of the proposed PRCEA

The pseudocode of PRCEA is outlined in **Algorithm 1**. It can be seen that PRCEA collaboratively evolves a main population $\mathbb{P}^{\text{main}}$ and an auxiliary population $\mathbb{P}^{\text{aux}}$ in a two-stage framework. During the initialization phase, a parent population $\mathbb{P}^{\text{main}}$ with N solutions is randomly generated in the decision space Ω and copied to $\mathbb{P}^{\text{aux}}$ (line 1), while the offspring population $\mathbb{O}$ is initialized as an empty set (line 2). Meanwhile, the switch flag *switch* is initialized to *false* (line

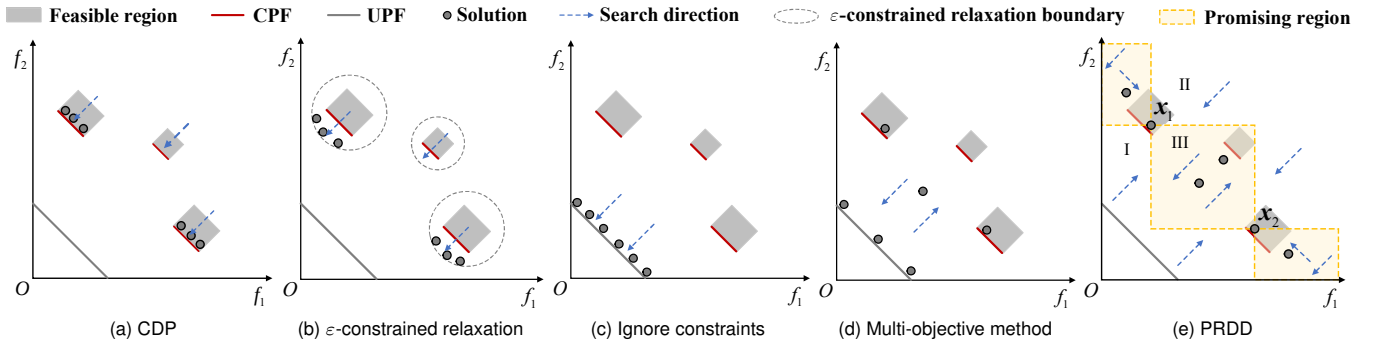


Fig. 1. Illustration of the effects of existing CHTs.

3), indicating that PRCEA starts in the first stage; when set to *true*, it signals the transition to the second stage.

Algorithm 1 Procedure of PRCEA

Input: Population size: N .

Output: The final main population: $\mathbb{P}^{\text{main}}$.

- 1: Initialize the main population $\mathbb{P}^{\text{main}}$ with N random solutions in Ω , and copy it to the auxiliary population $\mathbb{P}^{\text{aux}}$;
- 2: Initialize the offspring population: $\mathbb{O} \leftarrow \emptyset$;
- 3: Initialize the switch flag: *switch* $\leftarrow$ *false*;
- 4: **while** the stop criterion is not met **do**
- 5: $\mathbb{P}^{\text{main}} \leftarrow$ Retain up to N FNDSSs from $(\mathbb{P}^{\text{main}} \cup \mathbb{O})$ according to the CDP method;
- 6: $\mathbb{P}^{\text{aux}} \leftarrow$ Select N solutions from $(\mathbb{P}^{\text{aux}} \cup \mathbb{O})$ by the PRDD method; //Algorithm 2
- 7: Update *switch*;
- 8: **if** *switch* == *false* **then**
- 9: $\mathbb{O}_1 \leftarrow$ Generate offspring by executing the AES strategy on $\mathbb{P}^{\text{main}}$; //Algorithm 3
- 10: $\mathbb{O}_2 \leftarrow$ Generate offspring by executing the AES strategy on $\mathbb{P}^{\text{aux}}$; //Algorithm 3
- 11: **else**
- 12: $\mathbb{O}_1 \leftarrow$ Generate offspring by applying the neighbor pairing-based DE operator to $\mathbb{P}^{\text{main}}$;
- 13: $\mathbb{O}_2 \leftarrow$ Generate offspring by applying the neighbor pairing-based DE operator to $\mathbb{P}^{\text{aux}}$;
- 14: **end if**
- 15: $\mathbb{O} \leftarrow \mathbb{O}_1 \cup \mathbb{O}_2$;
- 16: **end while**
- 17: $\mathbb{P}^{\text{main}} \leftarrow$ Select up to N FNDSSs from $(\mathbb{P}^{\text{main}} \cup \mathbb{O})$ by the CDP-based method;

During the main loop, the first step is the population update (lines 5-6). Specifically, $\mathbb{P}^{\text{main}}$ is dedicated to CPF approximation and preserves only the FNDSSs determined by CDP. If the number of FNDSSs exceeds N , the truncation strategy proposed in SPEA2 [30] is invoked to prune $\mathbb{P}^{\text{main}}$ to the required size N . Conversely, $\mathbb{P}^{\text{aux}}$ is updated by the PRDD method, as detailed in Section III-B. In the first stage, PRDD performs global exploration over the entire search space to assist $\mathbb{P}^{\text{main}}$ in rapidly converging toward global optimal feasible regions. Once $\mathbb{P}^{\text{main}}$ is non-empty and its convergence stagnation is detected following the procedure in [36], it is assumed that $\mathbb{P}^{\text{main}}$ has reached the CPF. Consequently, *switch* is toggled

from *false* to *true*, and PRDD shifts its search focus to the promising region surrounding $\mathbb{P}^{\text{main}}$ to enhance local diversity and identify more feasible Pareto optimal solutions. Afterward, the proposed AES strategy and the neighbor pairing-based DE operator [45] are applied to generate offspring in the first and second stages, respectively. Notably, the former leverages a large search step size to facilitate global exploration, while the latter ensures fine-grained local exploitation of the CPF in the promising region. Detailed implementation of the AES strategy is provided in Section III-C. Finally, output $\mathbb{P}^{\text{main}}$ if the termination conditions are met (line 17).

B. Promising Region Detection and Diversity Enhancement

This section introduces the proposed PRDD method, which endeavors to evenly explore the promising region encompassing the CPF. The pseudocode of the PRDD method is presented in Algorithm 2. Fig. 2 illustrates the operational mechanism of PRDD through three representative examples, where x_1 to x_{16} are 16 solutions in the combined population $\mathbb{CP} (\mathbb{P}^{\text{aux}} \cup \mathbb{O})$, y_1 and y_2 are two FNDSSs in $\mathbb{P}^{\text{main}}$, and the corresponding ϕ values are listed in parentheses.

As shown in Algorithm 2, $\mathbb{CP}$ is first divided into three disjoint subsets: $\mathbb{PR}$, $\mathbb{PR}^>$, and $\mathbb{PR}^<$, aiming at the enhancement of diversity, feasibility, and convergence, respectively (lines 1-5). In the first stage, when $\mathbb{P}^{\text{main}}$ may be empty or far from the CPF, PRDD is responsible for uniformly exploring the entire search space to find constrained Pareto optimal solutions. Therefore, $\mathbb{PR} = \mathbb{CP}$, while both $\mathbb{PR}^>$ and $\mathbb{PR}^<$ are empty sets. In the second stage, with $\mathbb{P}^{\text{main}}$ having obtained a set of stably converged FNDSSs in close proximity to the CPF, their mutually non-dominated region is defined as the promising region that potentially contains the complete CPF. To provide sufficient expansion force for spreading the FNDSSs along the entire CPF, PRDD ends the global uniform search and focuses on uniformly searching the promising region. At this point, $\mathbb{P}^{\text{main}}$ is used to partition $\mathbb{CP}$, yielding:

$$\begin{cases} \mathbb{PR} = \{x \in \mathbb{CP} | \forall y \in \mathbb{P}^{\text{main}}, y \not\prec x \wedge x \not\prec y\}, \\ \mathbb{PR}^> = \{x \in \mathbb{CP} | \exists y \in \mathbb{P}^{\text{main}}, x \prec y\}, \\ \mathbb{PR}^< = \{x \in \mathbb{CP} | \exists y \in \mathbb{P}^{\text{main}}, y \prec x\}, \end{cases} \quad (6)$$

where $\mathbb{PR}$, $\mathbb{PR}^>$, and $\mathbb{PR}^<$ are the solution sets located in the promising region, the dominating region of $\mathbb{P}^{\text{main}}$, and the dominated region of $\mathbb{P}^{\text{main}}$, respectively. In Fig. 2,

Algorithm 2 The PRDD method

Input: Main population: $\mathbb{P}^{\text{main}}$; combined population of $\mathbb{P}^{\text{aux}}$ and $\mathbb{O}$: $\mathbb{CP}$; population size: N , switch flag: *switch*.
Output: Updated $\mathbb{P}^{\text{aux}}$.

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1: if switch == false then
2:    $\mathbb{PR} \leftarrow \mathbb{CP}$ ,  $\mathbb{PR}^{\succ} \leftarrow \emptyset$ ,  $\mathbb{PR}^{\prec} \leftarrow \emptyset$ ;
3: else
4:   Use  $\mathbb{P}^{\text{main}}$  to partition  $\mathbb{CP}$  into three exclusive subsets,
      $\mathbb{PR}$ ,  $\mathbb{PR}^{\succ}$ ,  $\mathbb{PR}^{\prec}$ , according to Eq. (6);
5: end if
6:  $\mathbb{P}^{\text{aux}} \leftarrow \emptyset$ ;
7: if  $|\mathbb{PR}| \geq N$  then
8:    $\mathbb{P}^{\text{aux}} \leftarrow$  Select  $N$  solutions from  $\mathbb{PR}$  based on EPD;
9: else
10:   $\mathbb{P}^{\text{aux}} \leftarrow \mathbb{P}^{\text{aux}} \cup \mathbb{PR}$ ;
11:  if  $|\mathbb{PR}^{\succ}| \geq N - |\mathbb{PR}|$  then
12:     $\mathbb{P}^{\text{aux}} \leftarrow \mathbb{P}^{\text{aux}} \cup \{\text{the top } (N - |\mathbb{PR}|) \text{ solutions in } \mathbb{PR}^{\succ} \text{ determined by ERPD}\}$ ;
13:  else
14:     $\mathbb{P}^{\text{aux}} \leftarrow \mathbb{P}^{\text{aux}} \cup \mathbb{PR}^{\succ}$ ;
15:     $\mathbb{P}^{\text{aux}} \leftarrow \mathbb{P}^{\text{aux}} \cup \{\text{the top } (N - |\mathbb{PR}| - |\mathbb{PR}^{\succ}|) \text{ solutions in } \mathbb{PR}^{\prec} \text{ selected based on PD}\}$ ;
16:  end if
17: end if

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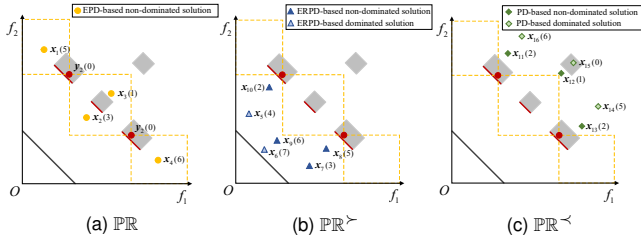


Fig. 2. Illustration of the benefits of the proposed PRDD method, where the gray regions are feasible, the promising region is enclosed by yellow dashed lines, and the black and red lines are the UPF and CPF, respectively.

$\mathbb{PR} = \{x_1, x_2, x_3, x_4\}$, $\mathbb{PR}^{\succ} = \{x_5, x_6, x_7, x_8, x_9, x_{10}\}$, and $\mathbb{PR}^{\prec} = \{x_{11}, x_{12}, x_{13}, x_{14}, x_{15}, x_{16}\}$.

Subsequently, the selection procedure is initiated (lines 7-17). PRDD assigns the highest priority to $\mathbb{PR}$, followed by $\mathbb{PR}^{\succ}$, and finally $\mathbb{PR}^{\prec}$. This is evidently true in the first stage, as only $\mathbb{PR}$ is non-empty. In the second stage, if all FNDSSs in $\mathbb{P}^{\text{main}}$ are Pareto optimal, the promising region must contain the complete CPF; otherwise, the CPF may reside in the dominating region of $\mathbb{P}^{\text{main}}$ [7], [45]. Given that $\mathbb{P}^{\text{main}}$ has already converged to the vicinity of the CPF in the first stage, the promising region is more likely to contain the CPF than the dominating region of $\mathbb{P}^{\text{main}}$. Moreover, the CPF cannot exist in the dominated region of $\mathbb{P}^{\text{main}}$. In the following, three specific selection mechanisms are sequentially applied to $\mathbb{PR}$, $\mathbb{PR}^{\succ}$, and $\mathbb{PR}^{\prec}$ to select N solutions for the next generation.

1) *Selecting Well-Distributed Solutions from $\mathbb{PR}$* : If $|\mathbb{PR}| \geq N$, then $\mathbb{P}^{\text{aux}}$ will be selected from $\mathbb{PR}$ (lines 4-5). To enhance search diversity of $\mathbb{PR}$, we define a new dominance relation called EPD, which is essentially the classical PD applied to the augmented objective space of MOP (3). Firstly, all individuals

in $\mathbb{PR}$ undergo the EPD-based non-dominated sorting and are assigned fitness values according to their dominance relationships and density information as in SPEA2. Afterwards, the truncation strategy of SPEA2 is applied to prune $\mathbb{PR}$ to a size of N . In Fig. 2(a), the EPD-based non-dominated solutions (yellow circles) are well-dispersed and exhibit either good objective values or low ϕ values. By prioritizing these solutions, individuals near or inside the detected or undetected feasible regions can be preserved, thereby facilitating both exploitation of known feasible regions and exploration of new ones.

2) *Selecting Solutions Near the CPF from $\mathbb{PR}^{\succ}$* : If $|\mathbb{PR}| < N$ and $|\mathbb{PR}^{\succ}| \geq N - |\mathbb{PR}|$, then $\mathbb{PR}$ is combined with the $(N - |\mathbb{PR}|)$ solutions selected from $\mathbb{PR}^{\succ}$ to form $\mathbb{P}^{\text{aux}}$. $\mathbb{PR}^{\succ}$ follows the same selection mechanism as $\mathbb{PR}$, except that the dominance relationships among individuals in $\mathbb{PR}^{\succ}$ are evaluated by a new dominance relation termed ERPD, which is defined as the classical PD applied to the following MOP:

$$\min \mathbf{f}(\mathbf{x}) = (-f_1(\mathbf{x}), -f_2(\mathbf{x}), \dots, -f_m(\mathbf{x}), \phi(\mathbf{x}))^T, \mathbf{x} \in \Omega. \quad (7)$$

MOP (7) implies two ideas: 1) maximization of the original objective functions; and 2) minimization of the overall constraint violation. From Fig. 2(b), the ERPD-based non-dominated solutions are x_7 to x_{10} , which either have low ϕ values or are near the promising region with high diversity. Therefore, preserving these infeasible solutions can accelerate convergence toward $\mathbb{P}^{\text{main}}$ from the infeasible side along diverse search directions, and help discover new feasible regions. If only the maximization of the original objective functions is considered, then the non-dominated solutions are x_8 to x_{10} . As a result, the solution x_7 , which has a lower ϕ value than x_8 and might be close to an undetected feasible region, risks being eliminated. By contrast, if only the minimization of $\phi(\mathbf{x})$ is taken into account, then the solutions x_6 and x_9 near the middle feasible region become less competitive, resulting in a loss of population diversity. Furthermore, due to the presence of deceptive constraints, a solution closer to the feasible region may have a larger ϕ value [45]. For instance, x_8 has a higher ϕ value than x_7 . If x_7 is retained while x_8 is removed, the evolutionary direction may deviate from the direction toward the feasible region.

3) *Selecting Well-Converged Solutions from $\mathbb{PR}^{\prec}$* : If $|\mathbb{PR}| < N$ and $|\mathbb{PR}^{\succ}| < N - |\mathbb{PR}|$, which means that $\mathbb{PR} \cup \mathbb{PR}^{\succ}$ is insufficient to fill $\mathbb{P}^{\text{aux}}$, then the remaining $(N - |\mathbb{PR}| - |\mathbb{PR}^{\succ}|)$ will be chosen from $\mathbb{PR}^{\prec}$ based solely on convergence without reference to feasibility. Specifically, $\mathbb{PR}^{\prec}$ adopts the PD-based selection mechanism, thereby guiding solutions toward the CPF along diverse search directions. As shown in Fig. 2(c), the non-dominated solutions x_{11} , x_{12} , and x_{13} are favored over the dominated solutions x_{14} , x_{15} , and x_{16} . In fact, the individuals in $\mathbb{PR}^{\prec}$ do not require the expenditure of computational resources to pursue constraint satisfaction. For example, if x_{15} is accepted while x_{12} is discarded for better constraint satisfaction, the evolutionary direction may be contrary to the desired direction of convergence toward the CPF.

To summarize, EPD promotes a diverse distribution of $\mathbb{PR}$;

ERPD guides $\mathbb{P}^{\rightarrow}$ to approach the promising region from the infeasible side while detecting new feasible regions; and PD accelerates the convergence of $\mathbb{P}^{\leftarrow}$ to the CPF. By assigning descending priority to $\mathbb{P}^{\rightarrow}$, $\mathbb{P}^{\rightarrow}$, and $\mathbb{P}^{\leftarrow}$, the proposed PRDD can effectively detect the promising region and focus the search effort on it without local crowding, thereby capturing the entire CPF.

C. Accelerated Evolutionary Search for Reproduction

To enhance search efficiency in large-scale decision spaces and accelerate convergence, we propose a novel AES strategy that incorporates inter-learning and intra-learning mechanisms. The pseudocode of AES is presented in **Algorithm 3**, which mainly consists of the following four components:

Algorithm 3 Reproduction

Input: Parent population: $\mathbb{P}$.

Output: Offspring population $\mathbb{O}$.

```

1:  $\mathbb{O} \leftarrow \emptyset$ ;
2: Normalize the objective vectors in  $\mathbb{P}$  using Eq. (8);
3:  $\mathbb{S}_e$  and  $\mathbb{S}_p \leftarrow$  Divide  $\mathbb{P}$  into two subsets;
4: /* Inter-learning
5: for each  $x \in \mathbb{S}_p$  do
6:    $x^l \leftarrow$  Select the solution in  $\mathbb{S}_e$  that has the smallest
     angular distance to  $x$  based on Eq. (9);
7:    $x^c \leftarrow$  Generate a child solution by Eq. (11);
8:   Repair  $x^c$  by Eq. (15);
9:    $\mathbb{O} \leftarrow \mathbb{O} \cup \{x^c\}$ ;
10: end for
11: /* Intra-learning
12: for each  $x \in \mathbb{S}_e$  do
13:    $x^{c,1}, x^{c,2}, x^{c,3}, x^{c,4} \leftarrow$  Generate four child solutions
     by Eq. (16);
14:   Repair  $x^{c,1}, x^{c,2}, x^{c,3}, x^{c,4}$  by Eq. (15);
15:    $\mathbb{O} \leftarrow \mathbb{O} \cup \{x^{c,1}, x^{c,2}, x^{c,3}, x^{c,4}\}$ ;
16: end for
```

1) *Normalization in Line 2:* To eliminate the scale differences across different objectives, the objective vector $f(x)$ of each solution $x \in \mathbb{P}$ is normalized to $f'(x) = (f'_1(x), f'_2(x), \dots, f'_m(x))^T$ using the following formula:

$$f'_k(x) = \frac{f_k(x) - z_k^{\min}}{z_k^{\max} - z_k^{\min}}, \quad k = 1, \dots, m \quad (8)$$

where $z_k^{\max}$ and $z_k^{\min}$ denote the maximum and minimum values of the k^{th} objective function f_k across all solutions in $\mathbb{P}$, respectively [19], [26]. The current population-based estimation is computationally efficient and easy to implement. It mitigates the normalization range compression that often occurs since the estimated minimum objective vector derived from all historical solutions is highly likely to fall far below the true CPF, while preserving valuable guidance from promising infeasible solutions that would otherwise be lost if only feasible solutions found so far are used.

2) *Clustering the Population $\mathbb{P}$ in Line 3:* $\mathbb{P}$ is divided into two distinct subsets, $\mathbb{S}_e$ and $\mathbb{S}_p$, where the solutions in $\mathbb{S}_e$ outperform those in $\mathbb{S}_p$ in terms of convergence, diversity, or

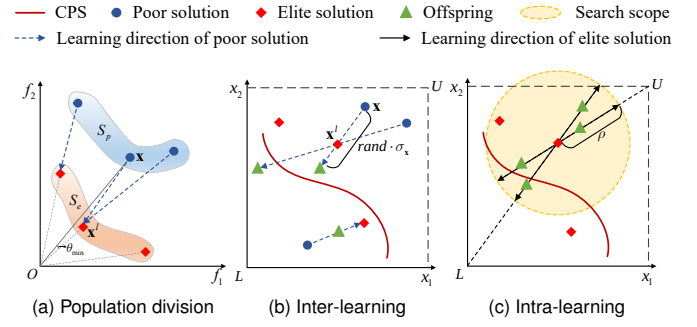


Fig. 3. Illustration of the AES strategy for reproduction.

feasibility. In particular, the solutions in $\mathbb{P}$ are first sorted in ascending order according to their fitness values. Thereafter, the top half of the ranked solutions are allocated to $\mathbb{S}_e$ and the rest to $\mathbb{S}_p$, as a simple example shown in Fig. 3(a). Notably, the fitness values of the mating parents from $\mathbb{P}^{\text{main}}$ and $\mathbb{P}^{\text{aux}}$ are determined by the CDP-based method and PRDD method, respectively, which can be directly obtained from the population update process shown in lines 5-6 of **Algorithm 1**.

3) *Inter-Learning in Lines 5-9:* The solutions in $\mathbb{S}_p$ are typically far from the feasible regions and exhibit poor convergence and diversity. To this end, this paper proposes an inter-learning strategy, which guides each solution $x \in \mathbb{S}_p$ to learn from an elite solution $x^l \in \mathbb{S}_e$ that is most likely to guide x toward rapid convergence. Fig. 3(b) depicts the inter-learning diagram. Concretely, x^l shares the most similar search direction with x and is obtained as follows:

$$x^l = y : \operatorname{argmin}_{y \in \mathbb{S}_e} \theta(x, y), \quad (9)$$

$$\theta(x, y) = \arccos \left| \frac{f'(x)^T f'(y)}{\|f'(x)\| \cdot \|f'(y)\|} \right|, \quad (10)$$

where $\|\cdot\|$ is the ℓ_2 -norm, and $\theta(x, y)$ is the acute angle between the normalized objective vectors $f'(x)$ and $f'(y)$.

Having obtained x^l , a child solution is produced as follows:

$$x^c = x + \text{rand} \cdot \sigma_x \cdot \frac{x^l - x}{\|x^l - x\|}, \quad (11)$$

where rand is a random number sampled from $[0, 1]$, and σ_x is a customized scaling factor for x , calculated as follows:

$$\sigma_x = \sigma_x^{\min} + (\sigma_x^{\max} - \sigma_x^{\min}) \left(1 - \frac{FES}{\text{MaxFES}}\right)^2, \quad (12)$$

$$\sigma_x^{\min} = \|x^l - x\|, \quad (13)$$

$$\sigma_x^{\max} = \|u - l\|. \quad (14)$$

FES and MaxFES are the current and maximum number of function evaluations, respectively. $\sigma_x^{\min}$ and $\sigma_x^{\max}$ denote the minimum and maximum values of σ_x , respectively. u and l represent the upper and lower bound vectors of the decision space Ω , respectively. As indicated by Eqs. (12) and (13), both σ_x and $\sigma_x^{\min}$ are solution-dependent, i.e., they vary with x . Moreover, σ_x dynamically decreases from $\sigma_x^{\max}$ to $\sigma_x^{\min}$. Early in evolution, when $\mathbb{S}_e$ is often far from the CPF, a large σ_x empowers x to produce offspring that surpass x^l along the

search direction, thereby accelerating the exploration of the CPF. While at the late stage, when $\mathbb{S}_e$ closely approximates the CPF, a small σ_x allows x to search around x^l , ensuring effective exploitation of the CPF.

Before x^c is added to the offspring population $\mathbb{O}$, each of its components is repaired by [31]

$$x_k^c = \min(\max(x_k^c, l_k), u_k), \quad k = 1, \dots, n. \quad (15)$$

This ensures that x^c lies within the bounds of Ω .

4) *Intra-Learning in Lines 11-14*: The solutions in $\mathbb{S}_e$ may lie in the feasible regions, promising region, or have good objective values. Therefore, the primary task of $\mathbb{S}_e$ is to maintain diversity to escape local optima and cover all tiny/discrete CPF segments. In view of this, an intra-learning strategy is customized for $\mathbb{S}_e$ and its schematic diagram is provided in Fig. 3(c). Specifically, for each $x \in \mathbb{S}_e$, four new solutions are generated by perturbing x along directions toward or away from the lower and upper bounds (l and u):

$$\begin{aligned} x^{c,1} &= x + rand_1 \cdot \rho \cdot \frac{x - l}{\|x - l\|}, \\ x^{c,2} &= x + rand_2 \cdot \rho \cdot \frac{l - x}{\|l - x\|}, \\ x^{c,3} &= x + rand_3 \cdot \rho \cdot \frac{x - u}{\|x - u\|}, \\ x^{c,4} &= x + rand_4 \cdot \rho \cdot \frac{u - x}{\|u - x\|}, \end{aligned} \quad (16)$$

where $rand_1$ to $rand_4$ are four random values sampled from $[0, 1]$. ρ is a dynamic scaling factor that controls the search range, defined as:

$$\rho = \|u - l\| \cdot \left(1 - \frac{FES}{MaxFES}\right)^2. \quad (17)$$

At early evolutionary stage, $\rho \approx \|u - l\|$, allowing the algorithm to extensively explore the entire decision space to escape from local optima and rapidly converge to the vicinity of global optimal regions. As shown in Fig. 3(c), some offspring are close to the CPS. As the evolution proceeds, ρ gradually decreases, prompting the algorithm to conduct fine-grained exploitation around the CPS, thereby approximating the Pareto optimal solutions more efficiently.

Subsequently, $x^{c,1}, x^{c,2}, x^{c,3}, x^{c,4}$ are repaired by Eq. (15) and added to $\mathbb{O}$.

Discussion: The proposed AES provides comparative advantages over some state-of-the-art methods:

- *Comparison with CSOs [44], [47]*: Similar to AES, CSOs guide losers to learn from winners, yet most of them neglect the improvement of winners, which typically reside near feasible boundaries and possess a higher probability of intersecting with the CPS. By contrast, AES facilitates the local exploitation of winners through an intra-learning strategy with a dynamic search radius, which empowers the population to escape local optima and converge to better feasible regions.
- *Comparison with innovization progress operators [19], [26]*: These methods, mainly designed for unconstrained large-scale MOPs, typically train a machine learning

model using loser-winner pairs to learn the possible fastest convergent direction vector for each solution. However, they suffer from several critical limitations: 1) model training incurs huge extra computation cost and is susceptible to mode collapse [48]; 2) since winners and losers often reside in distinct regions (e.g., opposite sides of the CPF), a shared model may inadvertently push winners away from the feasible boundary. In contrast, AES is model-free and simple to implement. It uses an inter-learning strategy to guide losers toward rapid convergence along directions derived from their paired winners, while utilizing an intra-learning strategy to further advance winners.

D. Computational Complexity Analysis

PRCEA co-evolves a main population $\mathbb{P}^{\text{main}}$ and an auxiliary population $\mathbb{P}^{\text{aux}}$, both of them are assumed to have the same size N . When solving an LSCMOP with m objectives and n variables, the computational complexity of PRCEA per generation is primarily attributed to population update and offspring reproduction. During the process of population update, $\mathbb{P}^{\text{main}}$ first determine the dominance relations among individuals by the CDP method, incurring a total computational cost of $\mathcal{O}(mN^2)$. Then, the fitness assignment and truncation operations proposed in SPEA2 [30] are applied to prune $\mathbb{P}^{\text{main}}$ to the predefined size N , with an overall computational complexity of $\mathcal{O}(N^2 \log N)$. Meanwhile, $\mathbb{P}^{\text{aux}}$ is updated by the proposed PRDD method, in which both the population division and the three types of non-dominated sorting schemes have a worst-case computational complexity of $\mathcal{O}(mN^2)$, while the truncation operation entails $\mathcal{O}(N^2 \log N)$ computations. As for reproduction, the complexity of the proposed AES strategy mainly arises from the inter-learning and intra-learning components, which require $\mathcal{O}(mN^2)$ and $\mathcal{O}(nN)$ operations, respectively. Moreover, the complexity of the neighbor pairing-based DE operator is $\mathcal{O}(mN^2)$. Therefore, the overall complexity of PRCEA in one generation is $\{\mathcal{O}(mN^2) + \mathcal{O}(N^2 \log N)\}$.

IV. EXPERIMENTAL DESIGN

A. Benchmark Problems

To systematically test the performance of various algorithms, two types of test suites with dimensions ranging from 100 to 1000 are selected. The first type is the DASCOP test suite [49], which constructs constraints in the objective space and offers several parameter interfaces for users to adjust the number and size of feasible regions. The second type includes the recently proposed FCP [45], SDC [20] and LSCM [6] test suites, which contain only decision space constraints. Specifically, FCP contains deceptive constraints, and the randomly generated initial population resides in the infeasible region below the CPF, making the CPF difficult to locate through objective-only optimization. SDC allows users to adjust the feasibility difficulty, convergence difficulty, and the overlap degree between the CPF and UPF, thereby better testing the generalization performance of algorithms. LSCM incorporates characteristics from large-scale multi-objective optimization problems, CMOPs, and real-world problems,

aiming to provide a more realistic assessment of algorithm performance on LSCMOPs. Notably, the true CPF of each benchmark problem is available in its original paper.

B. Compared Algorithms and Their Parameter Settings

To verify the superiority of the proposed PRCEA in solving LSCMOPs, three CMOEAs (EMCMO [15], ICMA [45], and CMOES [46]) and four LSCMOEAs (ILCMO [7], CMOCSSO [42], LCMOEAs [19], and IMTCMO-BS [6]) are selected for performance comparison. Among them, EMCMO evolves two populations with and without constraints, and dynamically transfers parent or offspring solutions based on the tentative results. ICMA and CMOES respectively propose an indicator-based CHT and an even search strategy to guide the population to uniformly explore the promising region that are not dominated by any examined feasible solutions. The remaining four LSCMOEAs have been thoroughly reviewed in Section II-B. A summary of the search strategies and CHTs employed by these seven compared algorithms is provided in Table S-IX of the supplementary file.

The parameter settings of the above algorithms are consistent with those specified in their original papers. In the experiments, all algorithms are implemented on the PlatEMO platform [50] and undergo 31 independent runs on each problem, with the population size fixed at 100. Besides, the maximum number of function evaluations is adopted as the termination criterion for all the algorithms, which is set to $5000 \times n$ for all large-scale test problems to ensure a fair comparison under a consistent computational budget [6].

C. Performance Metrics

To quantitatively evaluate and compare the convergence and diversity of the final solution sets obtained by different algorithms, inverted generational distance (IGD) [51], inverted generational distance based on modified distance calculation (IGD^+) [52], and hypervolume (HV) [53] are selected as performance metrics. In particular, IGD measures the average distance from the reference point set to the obtained solution set. IGD^+ improves IGD by incorporating the dominance relation between a reference point and an objective vector into the distance measurement to ensure weak Pareto compliance. For IGD and IGD^+ calculation, 10000 reference points are uniformly generated on the true CPF of each benchmark problem using the methods presented in [31] and [50]. HV quantifies the volume enclosed by the obtained solution set and a reference point $\mathbf{r} = (r, \dots, r)^T$ in the normalized objective space. As suggested in [54], r is set to 1.01 for bi-objective problems and 1.1 otherwise. Before HV calculation, the obtained solutions are normalized based on the ideal and nadir points derived from the true CPF. Smaller IGD/ IGD^+ values and larger HV values are desirable.

In the following experiments, the symbols “+”, “=”, and “-” denote superior, equivalent, and inferior performance of the compared algorithm relative to the proposed PRCEA, respectively, according to the Wilcoxon rank-sum test at the 0.05 significance level. The rankings of the algorithms are obtained via the Friedman test. “NaN” indicates that no

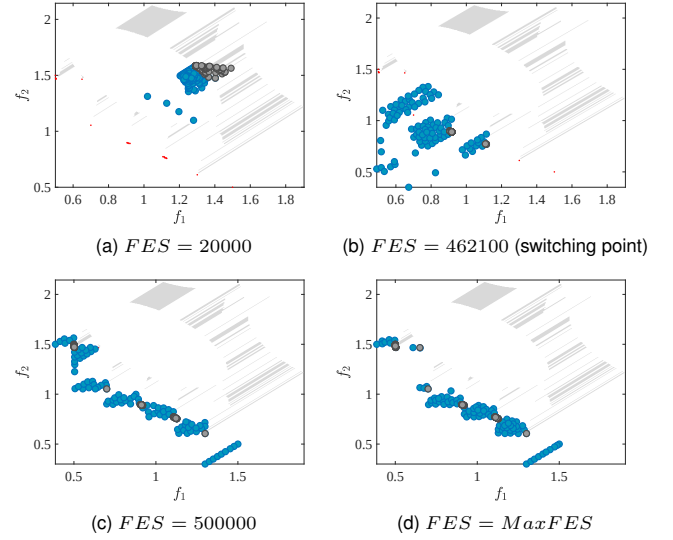


Fig. 4. The true distributions of $\mathbb{P}^{\text{main}}$ and $\mathbb{P}^{\text{aux}}$ on DASCMP6 at different budgets, where gray regions are feasible, the red line is the CPF, and the gray and blue circles are solutions in $\mathbb{P}^{\text{main}}$ and $\mathbb{P}^{\text{aux}}$, respectively.

feasible solution is obtained over 31 runs. The best result for each instance is highlighted in gray.

V. ANALYSIS OF PRCEA

A. Search Behavior Studies

To gain insight into the search behavior of PRCEA, this part visualizes its evolutionary process on DASCMP6 with $n = 500$. DASCMP6 has large infeasible regions, as well as tiny and disconnected feasible regions, making it difficult for algorithms to find the complete CPF. Fig. 4 depicts the true distributions of $\mathbb{P}^{\text{main}}$ (gray circles) and $\mathbb{P}^{\text{aux}}$ (blue circles) at different budgets in a single run, which has the median IGD value among 31 runs.

As shown in Fig. 4, at $FES = 20000$, $\mathbb{P}^{\text{main}}$ is trapped in external feasible regions because the CDP method places greater emphasis on constraints, while $\mathbb{P}^{\text{aux}}$ spreads across large infeasible regions with better objective values, and some solutions reach the boundaries of the global optimal feasible regions. With the assistance of $\mathbb{P}^{\text{aux}}$, $\mathbb{P}^{\text{main}}$ manages to cross large infeasible regions and converged to local CPF, as Fig. 4(b) shows. At this point, the switching condition is satisfied, triggering PRCEA to enter the second stage and construct the promising region. Then, at $FES = 500000$, $\mathbb{P}^{\text{aux}}$ is well-distributed in the promising region, as the proposed PRDD method can push the population into the promising region and facilitate extensive exploration, thereby discovering more Pareto optimal solutions. Finally, the distribution of $\mathbb{P}^{\text{main}}$ is significantly improved. From the above observations, it can be inferred that the proposed PRCEA can continuously leverage promising infeasible information to help detect new feasible regions in large-scale decision spaces.

B. Generalization Performance Studies

To evaluate the generalization performance of PRCEA in finding the complete CPF, five variants, named DASCMP6-1

to DASCMP6-5, are derived from DASCMP6 by modifying its parameter set $[\eta, \zeta, \gamma]$. In the objective space, these five functions differ in the number and shape of feasible regions, which gradually decrease in size. Moreover, 12 variants of SDC2 and SDC3 are created by adjusting the parameters HCT and b . Specifically, HCT denotes the proportion of constraint-related variables; a higher HCT value indicates a greater number of constraint-related variables. b controls the overlap degree between the CPF and UPF; the larger the value of b , the smaller the overlap between them. Table S-I of the supplementary file provides the parameter configurations for all variant functions. The mean IGD values for the eight algorithms on these variants at $n = 500$ are showcased via bar charts in Figs. S-8 to S-10 of the supplementary file, with the corresponding numerical results detailed in Table S-II of the supplementary file.

1) *The Impact of the Feasible Region Size:* According to Fig. S-8, the performance of ICMA, CMOCSSO, IMTCMOBS, and ILCMO deteriorates significantly as the feasible region size of DASCMP6 decreases. In contrast, CMOES, LCMOE, and PRCEA show only slight performance degradation trends. EMCMP remains stable on the former four problems, but its IGD value undergoes a sharp increase on DASCMP6-5. Based on Table S-II, PRCEA is inferior to CMOCSSO and ILCMO on DASCMP3-1, yet shows pronounced superiority on the remaining problems. The reason may be that when the feasible regions become very small, the constraint relaxation boundaries in CMOCSSO and ILCMO prematurely shrink to nearly zero before some tiny and isolated feasible regions are detected. Conversely, PRCEA can continuously exploit the promising region that contains the CPF, thereby finding more Pareto optimal solutions. Consequently, PRCEA can expand the search scope to effectively utilize infeasible information and exert its advantages more stably on LSCMPs with different sizes of feasible regions.

2) *The Impact of Scalable Constraint-Related Variables:* Similar observations can be drawn from both Figs. S-9(a) and S-10(a), where the IGD values of EMCMP, ICMA, CMOES, CMOCSSO, and ILCMO increase with HCT . In contrast, the results of LCMOE, IMTCMOBS and PRCEA show no remarkable change, with PRCEA consistently yielding the lowest IGD values. The Wilcoxon rank-sum test results in Table S-II indicate that PRCEA achieves the best results on four out of six variant functions. Therefore, it can be concluded that PRCEA is not significantly influenced by HCT and can effectively handle LSCMPs with varying numbers of constraint-related variables.

3) *The Impact of the Overlap Degree between the CPF and UPF:* The relative position of the CPF to the UPF influences algorithm performance around the CPF. As shown in Fig. S-9(b), the IGD values of EMCMP, ICMA, CMOES, CMOCSSO, LCMOE, and ILCMO increase markedly as the overlap decreases. Conversely, PRCEA and IMTCMOBS show similar and milder degradation, with PRCEA retaining a lower IGD level. In Fig. S-10(b), all algorithms maintain stable performance except LCMOE, which experiences a notable rise in IGD value at $b = 1.9$. Specifically, PRCEA achieves three best IGD values, validating its superior stability

TABLE I
THE WILCOXON TEST RESULTS AND FRIEDMAN RANKINGS OF PRCEA
USING DIFFERENT METHODS FOR UPDATING $\mathbb{P}^{\text{aux}}$.

Vs. PRCEA	IGD		IGD ⁺		HV	
	+/-	Ranking	+/-	Ranking	+/-	Ranking
PRCEA-CDP	2/21/18	4.5854	2/21/18	4.5976	3/13/25	4.0732
PRCEA-PD	4/17/20	4.0854	5/20/16	4.0122	5/15/21	3.8049
PRCEA-epsilon	4/18/19	3.8049	4/21/16	3.6463	6/12/23	3.3293
PRCEA-PRDD1	8/16/17	3.0854	6/17/18	3.2317	5/13/23	3.4512
PRCEA-PRDD2	2/12/27	3.3171	2/12/27	3.3293	2/8/31	3.5122
PRCEA	/	2.122	/	2.1829	/	2.8293

and effectiveness in handling different degrees of separation between the UPF and CPF.

C. Verification of the Main Strategies

1) *The Effectiveness of the PRDD Strategy:* The proposed PRDD strategy aims to drive $\mathbb{P}^{\text{aux}}$ to intensively exploit the promising region, thereby assisting $\mathbb{P}^{\text{main}}$ in achieving more comprehensive and diverse coverage of the CPF. To evaluate the effectiveness of PRDD, we create five variants labeled as PRCEA-CDP, PRCEA-PD, PRCEA- ϵ , PRCEA-PRDD1, and PRCEA-PRDD2. In the first three, $\mathbb{P}^{\text{aux}}$ adopts the environmental selection of SPEA2, with dominance relationships determined by CDP, PD, and the ϵ -method [42], respectively. PRCEA-PRDD1 and PRCEA-PRDD2 adopt the $\mathbb{CP}$ partitioning strategies from the first and second stages of **Algorithm 2**, respectively. Notably, PRCEA-PRDD2 defaults to the first-stage partitioning strategy if $\mathbb{P}^{\text{main}}$ is empty.

Tables S-III to S-V of the supplementary file report the metric results for benchmark functions with $n = 500$, and a brief summary is presented in Table I. With respect to the IGD metric, PRCEA-CDP, PRCEA-PD, PRCEA- ϵ , PRCEA-PRDD1, and PRCEA-PRDD2 outperform PRCEA on 2, 4, 4, 8, and 2 test functions, respectively, but yield inferior results on 21, 17, 18, 16, and 12 functions. Similar observations are found in the IGD⁺ and HV results. This may be attributed to the following factors: CDP prioritizes feasibility, thus failing to help $\mathbb{P}^{\text{main}}$ escape local optima; PD pushes $\mathbb{P}^{\text{aux}}$ toward the UPF, yet offers limited support to $\mathbb{P}^{\text{main}}$ when the CPF separates from the UPF; the relaxed constraint boundary of the ϵ method gradually shrinks to zero, making it difficult for solutions to spread along the CPF in the late evolutionary phase; PRCEA-PRDD1 continuously utilizes EPD to promote a well-distributed $\mathbb{P}^{\text{aux}}$, leaving few computational resources for exploiting regions around the CPF; and PRCEA-PRDD2 may waste excessive computational resources exploring feasible solutions far from the CPF. In contrast, these shortcomings are effectively addressed by the proposed PRDD method. By first seeking a set of convergence-stable FNDs and then focusing search efforts on the promising region defined by them, PRDD maintains a sustained expansion force to diffuse solutions into other undetected global optimal feasible regions. According to the Friedman test results, PRCEA ranks first in terms of all three metrics. Therefore, the effectiveness of PRDD is empirically validated.

2) *The Effectiveness of the AES Strategy:* In this paper, the proposed AES strategy and the neighbor pairing-based DE

TABLE II
THE WILCOXON TEST RESULTS AND FRIEDMAN RANKINGS OF PRCEA
USING DIFFERENT REPRODUCTION STRATEGIES.

Vs. PRCEA	IGD		IGD ⁺		HV	
	+/-/=	Ranking	+/-/=	Ranking	+/-/=	Ranking
PRCEA-GA	3/33/5	4.2439	4/33/4	4.2683	5/27/9	4.3171
PRCEA-DE	2/29/10	4.8293	2/29/10	4.8537	1/27/13	4.622
PRCEA-AES	2/37/2	4.3902	2/37/2	4.3902	1/32/8	4.3415
PRCEA-inter	4/26/11	3.6098	5/25/11	3.561	4/23/14	3.5
PRCEA-intra	5/9/27	2.7634	5/12/24	2.4756	5/13/23	2.5976
PRCEA	/	1.9634	/	1.9512	/	2.122

operator are alternately used to generate offspring. To verify the efficiency of this idea, we design five variants: PRCEA-GA, PRCEA-DE, PRCEA-AES, PRCEA-inter, and PRCEA-intra. Specifically, PRCEA-GA replaces the hybrid mechanism with standard SBX and PM operators, while PRCEA-DE and PRCEA-AES rely exclusively on the neighbor pairing-based DE operator and the AES strategy, respectively. PRCEA-inter and PRCEA-intra retain the inter-learning and intra-learning components of AES, respectively. The metric results obtained from solving the four test suites with $n = 500$ are detailed in Tables S-VI to S-VIII of the supplementary file and summarized in Table II. Additionally, the distribution of offspring populations generated by PRCEA and its variants on FCP1 is visualized in Fig. S-11.

Clearly, PRCEA, PRCEA-inter, and PRCEA-intra achieve better metric rankings than PRCEA-DE and PRCEA-GA, indicating that search efficiency can be improved by AES and its two components: inter-learning and intra-learning. In Fig. S-11, PRCEA-DE and PRCEA-GA fail to find feasible solutions because their generated offspring remain clustered tightly around their parents, resulting in limited exploration capability. In contrast, the offspring of the remaining algorithms are widely distributed across the objective space, enabling them to successfully identify feasible solutions. Specifically, PRCEA-inter and PRCEA-intra exhibit superior convergence and diversity, respectively, while PRCEA achieves the best balance of both rapid convergence and high diversity, affirming the necessity of synergistically utilizing the inter-learning and intra-learning mechanisms to evolve both loser and winner particles. However, despite consistently generating offspring using AES, PRCEA-AES is inferior to PRCEA on 37, 37, and 32 out of 41 functions in terms of IGD, IGD⁺ and HV, respectively, and even fails to find feasible solutions on certain test problems, such as SCD14 and LSCM11. This is because AES focuses more on accelerating convergence at the expense of overall diversity by using larger search steps, which can be compensated by the DE operator that operates with smaller search steps. PRCEA achieves the best rankings on both metrics, as AES can quickly generate offspring near the optimal solutions, while DE enables finer adjustments to all decision variables and therefore slowly approach the optimal value for each single variable. Therefore, the alternating use of AES and DE effectively balances convergence and diversity.

TABLE III
A SUMMARY OF THE COMPARATIVE RESULTS.

IGD (+/-/=)								
Vs. PRCEA on	n	EMCMO	ICMA	CMOES	CMOCSO	LCMOEA	IMTCMO-BS	ILCMO
DASCMP	100	0/6/3	0/9/0	0/6/3	0/9/0	0/8/1	1/6/2	1/8/0
	500	1/5/3	0/9/0	1/6/2	0/9/0	0/8/1	0/6/3	1/8/0
	1000	0/5/4	0/9/0	0/6/3	0/7/2	0/9/0	1/6/2	1/7/1
FCP	100	0/5/0	1/4/0	0/5/0	0/5/0	0/5/0	0/4/1	1/4/0
	500	0/4/1	1/4/0	0/4/1	0/4/1	0/4/1	1/3/1	0/4/1
	1000	0/4/1	1/4/0	0/4/1	0/4/1	0/4/1	0/2/3	0/4/1
SDC	100	1/12/2	0/12/3	0/11/4	0/14/1	0/15/0	0/2/13	0/10/5
	500	1/13/1	0/14/1	0/13/2	0/14/1	0/15/0	2/7/6	3/10/2
	1000	0/13/2	0/13/2	0/15/0	1/14/0	0/15/0	3/8/4	4/8/3
LSCM	100	0/12/0	0/9/3	0/12/0	0/12/0	1/8/3	1/1/10	2/2/8
	500	1/9/2	0/10/2	1/10/1	0/9/3	0/11/1	2/4/6	2/4/6
	1000	0/9/3	0/11/1	0/9/3	0/10/2	0/11/1	1/7/4	2/9/1
IGD ⁺ (+/-/=)								
Vs. PRCEA on	n	EMCMO	ICMA	CMOES	CMOCSO	LCMOEA	IMTCMO-BS	ILCMO
DASCMP	100	2/7/0	0/8/1	0/7/2	0/9/0	0/8/1	1/7/1	3/5/1
	500	1/5/3	0/9/0	1/7/1	0/9/0	1/8/0	0/8/1	3/5/1
	1000	1/5/3	0/9/0	0/7/2	0/8/1	0/9/0	0/8/1	2/4/3
FCP	100	0/5/0	1/4/0	0/4/1	0/5/0	0/5/0	0/4/1	1/4/0
	500	0/4/1	1/4/0	1/4/0	0/5/0	0/5/0	1/4/0	0/4/1
	1000	1/4/0	0/5/0	1/4/0	0/5/0	0/5/0	0/3/2	0/4/1
SDC	100	0/12/3	0/12/3	1/12/2	0/14/1	0/15/0	0/2/13	0/10/5
	500	0/13/2	0/14/1	0/13/2	0/14/1	0/15/0	2/7/6	3/10/2
	1000	0/13/2	0/13/2	0/15/0	1/14/0	0/15/0	3/8/4	4/8/3
LSCM	100	0/12/0	1/8/3	0/11/1	0/12/0	0/10/2	2/1/9	2/3/7
	500	1/9/2	0/10/2	1/10/1	0/9/3	0/11/1	2/4/6	2/4/6
	1000	0/9/3	0/11/1	0/9/3	0/10/2	0/11/1	1/6/5	2/9/1
HV (+/-/=)								
Vs. PRCEA on	n	EMCMO	ICMA	CMOES	CMOCSO	LCMOEA	IMTCMO-BS	ILCMO
DASCMP	100	3/6/0	0/7/2	2/5/2	0/9/0	0/9/0	1/6/2	3/6/0
	500	4/5/0	0/9/0	4/5/0	0/9/0	0/9/0	0/6/3	4/5/0
	1000	4/5/0	0/9/0	3/6/0	0/8/1	0/9/0	0/7/2	2/6/1
FCP	100	0/5/0	1/4/0	0/4/1	0/5/0	0/5/0	0/4/1	1/4/0
	500	0/4/1	1/4/0	1/4/0	0/5/0	0/5/0	0/3/2	0/4/1
	1000	1/4/0	0/5/0	1/4/0	0/5/0	0/5/0	0/3/2	0/4/1
SDC	100	1/13/1	0/14/1	0/13/2	0/15/0	0/15/0	0/3/12	0/12/3
	500	1/12/2	0/12/3	0/12/3	0/13/2	0/14/1	2/6/7	5/9/1
	1000	0/10/5	0/12/3	0/12/3	0/12/3	0/13/2	2/9/4	3/7/5
LSCM	100	0/11/1	0/8/4	0/11/1	0/11/1	1/8/3	0/11/1	2/2/8
	500	1/8/3	0/7/5	1/8/3	0/9/3	0/9/3	1/4/7	2/4/6
	1000	0/7/5	0/6/6	0/7/5	0/10/2	0/9/3	2/5/5	2/4/6

VI. COMPARATIVE EXPERIMENTAL RESULTS AND ANALYSIS

A. Comparison Studies on Benchmark Problems

In this section, PRCEA is compared with seven peer algorithms on the DASCMP, FCP, SDC, and LSCM test suites with dimensions ranging from 100 to 1000. The comparison results for the four test suites are presented in Tables S-X to S-XXI of the supplementary file and summarized in Table III. In addition, Fig. 5 plots the fixed target-based empirical cumulative distribution function (ECDF) curves [55] of the IGD⁺ metric to compare the convergence performance of different algorithms across four benchmarks. The target values are 50 log-linearly spaced points between the best and worst found IGD⁺ values per test instance. In any subfigure of Fig. 5, the ECDF curve of each algorithm aggregates the results from 31 independent runs on one test suite at dimension 500. Figs. S-12 to S-15 of the supplementary file provide the median IGD⁺ convergence curves for each test problem at $n = 500$. Since the performance analysis of the IGD, IGD⁺, and HV results is similar, we will focus on analyzing the IGD⁺ results to save space.

1) *DASCMP Problems*: According to the Wilcoxon test results shown in Table III, PRCEA exhibits a clear competitive edge over the seven peer algorithms. Specifically, ICMA, C-MOES, CMOCSO, LCMOEA, and IMTCMO-BS are superior to PRCEA on no more than one instance. EMCMO and

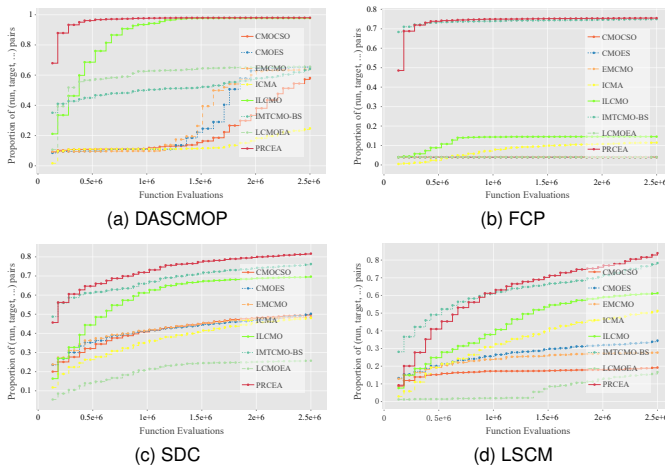


Fig. 5. Empirical cumulative distribution function (ECDF) curves of the IGD^+ metric for eight algorithms aggregated on different test suites.

ILCMO significantly outperform PRCEA on 4 and 8 problems, respectively; however, they perform worse than PRCEA on 17, and 14 problems, respectively. As shown in Fig. 5(a), the ECDF curve of PRCEA exhibits the steepest slope during the early evolutionary stage and consistently remains at the top, indicating that, on average, PRCEA finds high-quality solutions faster than other algorithms. Similar observations can be made in Fig. S-12, where PRCEA converges the fastest and achieves the best IGD^+ values on DASCMP1 and DASCMP3-6, while EMCMSO, ICMA, CMOES, CMOCSSO, and IMTCMO-BS fail to find feasible solutions on DASCMP4-8 until the late evolutionary stage. The superiority of PRCEA mainly stems from two aspects: 1) the proposed AES helps generate high-quality offspring to accelerate convergence to the CPF; and 2) the proposed PRDD provides sufficient expansion force to spread solutions from local CPF to other undetected global optimal feasible regions, thereby enhancing diversity.

2) *FCP Problems*: The initial population of FCP1-5 is situated in the region below the CPF, which is permeated by band-shaped local infeasible regions. Therefore, the algorithm should fully explore the objective space; otherwise, it may get trapped in local infeasible regions and fail to find any feasible solutions. As shown in Table S-XIV, only ILCMO and PRCEA are capable of finding feasible solutions for all problems, whereas the other algorithms encounter instances where no feasible solutions can be obtained. Particularly on FCP1-4, ICMA become ineffective when $n > 100$, EMCMSO, CMOES, CMOCSSO, and LCMOE fails to find feasible solutions across all three tested dimensions. In Fig. 5(b), IMTCMO-BS initially achieves the highest proportion of run-target pairs, indicating fast early convergence. However, the ECDF curve of PRCEA quickly surpasses that of IMTCMO-BS and remains the uppermost, as PRCEA shifts toward persistent exploitation around identified constrained Pareto optimal solutions to enhance local diversity, whereas the ϵ -constraint boundary of IMTCMO-BS prematurely shrinks to zero. The ECDF curves of the remaining algorithms consistently remain at lower positions. Consequently, these results validate the superiority of PRCEA in solving problems with deceptive constraints. The

ECDF curves of the remaining algorithms consistently stay at lower positions. Therefore, the superiority of PRCEA in handling deceptive constraints is clearly demonstrated.

3) *SDC Problems*: The 15 SDC problems impose constraints in the decision space, resulting in complex nonlinear relationships between objective optimization and constraint satisfaction, as well as objective spaces filled with local regions [20]. Consequently, algorithms are prone to getting trapped in local optima, especially in large-scale search spaces. As shown in Table III, PRCEA significantly outperforms EMCMSO, ICMA, CMOES, CMOCSSO, LCMOE, IMTCMO-BS, and ILCMO on 38, 39, 40, 42, 45, 17, and 28 out of 45 test problems, respectively. Besides, Fig. 5(c) shows that the ECDF curve of PRCEA remains the uppermost once FES reaches 250,000. This indicates that PRCEA possesses a superior average convergence speed and is competent to solve large-scale SDC problems with complex constraints.

However, PRCEA struggles with some problems. For example, it performs worse than IMTCMO-BS on SDC6, SDC11, and SDC13, and is outperformed by ILCMO on SDC9, SDC11, and SDC14. This is likely because $\mathbb{P}^{aux}$ of PRCEA uses the EPD-based CHT in the first stage, which prioritizes solutions with good objective values or lower ϕ values. Consequently, $\mathbb{P}^{aux}$ becomes widely distributed across the objective space and falls into the infeasible region located in the third quadrant, leaving limited computational resources to search for the CPF in the first quadrant. Conversely, IMTCMO-BS and ILCMO can search uniformly within the ϵ -relaxed constraint boundary, which enhances population diversity and prevents premature trapping in infeasible or local optimal regions. Moreover, PRCEA performs poorly on SDC2 and SDC10 with $n = 100$, but gradually gains superiority with increasing dimensionality and ultimately delivers the best results. SDC1, SDC4-5, SDC10, and SDC12 show pronounced multimodality due to the presence of numerous local regions in their objective spaces. Table S-XVII of the supplementary file reveals that all algorithms obtain worse IGD^+ values on these problems, probably because they all fall into local feasible regions far from the CPF. Compared to the seven competitors, PRCEA generally achieves significantly better results, which can be attributed to its utilization of a novel search method, AES, to accelerate convergence, and a uniform search strategy, PRDD, to consistently exploit the promising region. For the remaining SDC3, SDC7-8, and SDC15, no algorithm outperforms PRCEA, validating its effectiveness in solving problems with complex constraints.

4) *LSCM Problems*: The 12 LSCM problems are specifically designed for large-scale constrained multi-objective optimization, which also construct feasible regions in the decision space. From Table S-XX of the supplementary file, PRCEA ranks first on 20 out of the 36 test instances, achieving the highest winning frequency among the eight algorithms. In Fig. 5(d), PRCEA exhibits fast initial convergence and obtains a high proportion of the run-target pairs second only to IMTCMO-BS. However, after consuming approximately 10^6 function evaluations, the ECDF curve of PRCEA rises to the highest position, indicating that PRCEA can continuously improve its performance and find more high-quality solutions

during long-term optimization. Regarding the two types of compared algorithms, we have the following observations.

The three CMOEAs, EMCMO, ICMA, and CMOES, focus on designing effective CHTs to cope with constraints. Specifically, EMCMO is tailored for CMOPs involving objective-space constraints, where objectives and constraints exhibit simple linear relationships. It seeks to indirectly improve feasibility by optimizing a constraint-ignored auxiliary task. However, the objective space of LSCM problems is often filled with local regions. Therefore, overemphasis on objective optimization can lead the population to easily overlook feasible regions, making it more challenging to find feasible solutions for problems like LSCM8 and LSCM11-12, especially in high-dimensional search spaces. ICMA and CMOES respectively utilize an indicator-based CHT and an even-search strategy to uniformly explore the promising region not dominated by any examined feasible solutions. Consequently, they achieve better IGD^+ values than EMCMO on 25 and 23 problems, respectively, suggesting that uniform exploration of the objective space contributes to better performance on LSCM problems. Notably, ICMA yields better IGD^+ values than CMOES on 21 problems and successfully finds feasible solutions on more instances, as the DE operator is more suitable than the SBX and PM operators for managing the complex variable linkages in LSCM problems. However, EMCMO, ICMA, and CMOES perform worse than PRCEA on almost all LSCM problems, primarily because they lack effective mechanisms tailored for large-scale variables.

Similar to EMCMO, CMOCSSO, and LCMOEAs are designed for LSCMOPs with objective space constraints, aiming to reduce the ϕ values of solutions by optimizing the objective functions. Besides, they generate offspring with improved convergence and diversity through the paired CSO, CCSO, and learnable DE, respectively. Nevertheless, CMOCSSO, and LCMOEAs perform considerably worse than EMCMO and fail to obtain feasible solutions on many LSCM problems. This may be due to the relatively large search steps of their evolutionary operators, which impede the identification of small and disconnected feasible regions commonly present in LSCM problems. IMTCMO-BS, ILCMO, and PRCEA effectively mitigate this issue by alternating between traditional and novel evolutionary operators, and all are equipped with uniform search strategies. However, PRCEA generally gets the best IGD^+ results, particularly under higher-dimensional settings, as the relaxed constraint boundaries in IMTCMO-BS and ILCMO tend to shrink to zero before the full CPF is acquired, whereas the proposed PRDD enables PRCEA to continuously explore potential regions containing the CPF.

In conclusion, PRCEA emerges as the best solver and offers a promising approach for addressing various LSCMOPs. Due to space limitations, the statistical results for the four test suites based on the Wilcoxon rank-sum test and Friedman test are presented in Section S-V of the supplementary file. Besides, Figs. S-17 to S-21 of the supplementary file plot the final output populations obtained by the eight algorithms on DASCMP1, DASCMP3, SDC3, and LSCM5. The robustness tests on small-scale CMOPs are presented in Section S-VII of the supplementary file.

B. Application to the Coal Mine Integrated Energy System

This paper adopts the dispatch optimization model of CMIES from [4], which integrates the power grid, gas turbine (GT), photovoltaic (PV), wind turbine (WT), ventilation air methane (VAM) oxidation device (VOD), water source heat pump (WSHP), ground source heat pump (GSHP), air source heat pump (ASHP), absorption chiller (AC), electric chiller (EC), electric storage (ES), and thermal storage (TS), realizing coordination and energy conversion among the electricity, heat, and cooling subsystems. The model considers energy dispatch over one day and sets the time interval to one hour, resulting in a total of 24 periods and 432 decision variables. Additionally, the model minimizes the operating cost and the abandoned energy cost in a day, subject to constraints including the supply-demand balance of electricity, cooling, and thermal, GT ramp rate limits, and time-series constraints on storage devices (ES, TS). By setting different electrical, cooling, and thermal loads, along with predicted solar and wind power, 30 LSCMOPs termed CMIES1 to CMIES30 are created. Detailed load and predicted power data can be found in [4] and [3].

Since the CPF of each CMIES dispatch optimization problem is unknown, the FNDs of the combined solution set (including final solutions obtained by all algorithms) are used as the reference point set for IGD and IGD^+ calculations. For HV calculation, the ideal and nadir points used for normalization are estimated as the minimum and maximum objective vectors of this reference point set, respectively. The IGD^+ results are presented in Table IV, while the IGD and HV results are provided in Tables S-XXIII and S-XXIV of the supplementary file, respectively. It can be seen from Table IV that PRCEA significantly outperforms EMCMO, ICMA, CMOEA, CMOCSSO, LCMOEAs, and ILCMO across all CMIES problems. In addition, PRCEA is significantly inferior to IMTCMO-BS on CMIES2 and CMIES16, but it achieves statistically superior results on 19 out of the remaining 28 problems. To conclude, PRCEA has demonstrated its effectiveness on both benchmark functions (see Section VI-A) and practical applications. Therefore, PRCEA is recommended as the first-choice optimizer for addressing unknown LSCMOPs.

We provide a detailed analysis using CMIES1 as an example. Fig. 6(a) shows the IGD^+ convergence curves of eight algorithms with the median run, and the corresponding final FNDs are plotted in Fig. 6(b). As shown, IMTCMO-BS and PRCEA exhibit rapid convergence speeds at early evolutionary stage. However, when FES exceeds approximately 300,000, PRCEA consistently maintains the lowest IGD^+ curve and ultimately yields a population with the best convergence and diversity, validating the effectiveness of the proposed AES and PRDD in accelerating convergence and enhancing diversity, respectively. As demonstrated by the convergence results of the remaining CMIES dispatch problems in Fig. S-24, this observation is obtained across many CMIES dispatch problems, where some algorithms may outperform PRCEA at early evolution, yet PRCEA generally achieves the best final IGD^+ values. Furthermore, to verify whether the FNDs obtained by PRCEA meet practical production requirements, the left and right endpoint solutions of CMIES1 are selected

TABLE IV
IGD⁺ RESULTS OBTAINED BY EIGHT ALGORITHMS ON CMIES1 TO CMIES30.

Problem	EMCMO	ICMA	CMOES	CMOCSSO	LCMOEA	IMTCMO-BS	ILCMO	PRCEA
CMIES1	2.7277e+2 (7.61e+1) -	3.6492e+2 (1.20e+2) -	2.9349e+2 (8.02e+1) -	1.8886e+2 (4.30e+1) -	1.7662e+2 (1.42e+1) -	6.0149e+1 (1.33e+1) -	1.1419e+2 (1.69e+1) -	3.9992e+1 (1.59e+1)
CMIES2	3.5394e+2 (4.89e+1) -	6.1083e+2 (1.14e+2) -	4.1091e+2 (7.66e+1) -	3.0606e+2 (5.76e+1) -	2.7651e+2 (3.73e+1) -	1.0192e+2 (4.50e+1) +	3.0213e+2 (9.01e+1) -	1.5573e+2 (5.90e+1)
CMIES3	4.3244e+2 (7.34e+1) -	7.1345e+2 (8.53e+1) -	4.8946e+2 (5.52e+1) -	3.8112e+2 (6.17e+1) -	3.3310e+2 (4.22e+1) -	1.3860e+2 (3.82e+1) -	3.1793e+2 (6.97e+1) -	8.4065e+1 (3.12e+1)
CMIES4	4.4732e+2 (7.33e+1) -	6.7578e+2 (1.19e+2) -	4.7379e+2 (6.61e+1) -	3.9292e+2 (5.56e+1) -	3.2855e+2 (3.36e+1) -	1.2841e+2 (4.14e+1) =	3.5584e+2 (4.08e+1) -	1.2231e+2 (4.25e+1)
CMIES5	4.2039e+2 (6.27e+1) -	6.5153e+2 (1.50e+2) -	4.4225e+2 (5.60e+1) -	3.8601e+2 (4.70e+1) -	3.1002e+2 (4.14e+1) -	1.1404e+2 (4.64e+1) -	2.9304e+2 (7.73e+1) -	8.0919e+1 (3.99e+1)
CMIES6	4.5139e+2 (8.62e+1) -	6.6233e+2 (1.07e+2) -	4.9433e+2 (7.17e+1) -	4.0824e+2 (7.17e+1) -	3.2176e+2 (4.70e+1) -	1.2583e+2 (4.53e+1) -	3.2347e+2 (5.45e+1) -	9.0196e+1 (3.17e+1)
CMIES7	4.1263e+2 (6.15e+1) -	6.8632e+2 (1.23e+2) -	4.4903e+2 (4.57e+1) -	3.3762e+2 (6.23e+1) -	2.8917e+2 (3.05e+1) -	1.0405e+2 (3.36e+1) =	3.1792e+2 (4.82e+1) -	1.0290e+2 (5.02e+1)
CMIES8	4.2475e+2 (6.34e+1) -	6.1884e+2 (1.24e+2) -	4.5130e+2 (5.22e+1) -	3.6072e+2 (4.75e+1) -	3.0808e+2 (4.28e+1) -	1.3504e+2 (3.16e+1) -	3.3477e+2 (6.04e+1) -	7.8784e+1 (3.03e+1)
CMIES9	3.9835e+2 (5.51e+1) -	6.0211e+2 (1.04e+2) -	4.4529e+2 (5.20e+1) -	3.5990e+2 (7.71e+1) -	2.7165e+2 (4.94e+1) -	1.0018e+2 (3.47e+1) -	3.1124e+2 (7.69e+1) -	7.7592e+1 (5.69e+1)
CMIES10	4.5021e+2 (5.62e+1) -	6.6631e+2 (1.14e+2) -	5.0217e+2 (8.99e+1) -	4.2021e+2 (9.07e+1) -	3.3286e+2 (2.26e+1) -	1.4342e+2 (3.92e+1) -	3.2213e+2 (3.73e+1) -	9.2487e+1 (3.90e+1)
CMIES11	4.1546e+2 (8.53e+1) -	6.7928e+2 (1.13e+2) -	4.7686e+2 (7.45e+1) -	3.8943e+2 (6.15e+1) -	3.2377e+2 (6.36e+1) -	1.2645e+2 (4.70e+1) -	3.1975e+2 (7.43e+1) -	7.6836e+1 (3.76e+1)
CMIES12	4.7316e+2 (8.15e+1) -	6.9081e+2 (1.04e+2) -	4.7239e+2 (5.35e+1) -	3.8524e+2 (5.77e+1) -	3.2537e+2 (5.36e+1) -	1.1515e+2 (3.65e+1) -	3.2075e+2 (6.69e+1) -	8.3339e+1 (3.30e+1)
CMIES13	4.1821e+2 (7.96e+1) -	5.9298e+2 (8.75e+1) -	4.5876e+2 (5.91e+1) -	3.6895e+2 (6.16e+1) -	3.1216e+2 (2.16e+1) -	1.1471e+2 (3.61e+1) -	3.2824e+2 (4.47e+1) -	7.2984e+1 (3.18e+1)
CMIES14	4.4299e+2 (5.79e+1) -	6.6691e+2 (1.04e+2) -	4.6619e+2 (6.58e+1) -	3.7205e+2 (4.62e+1) -	3.0831e+2 (3.73e+1) -	1.2712e+2 (3.83e+1) -	3.0883e+2 (7.54e+1) -	1.0682e+2 (4.73e+1)
CMIES15	4.7025e+2 (8.62e+1) -	6.6158e+2 (1.33e+2) -	4.6436e+2 (6.22e+1) -	3.8935e+2 (5.25e+1) -	2.9630e+2 (3.29e+1) -	1.3624e+2 (3.84e+1) -	3.1305e+2 (4.78e+1) -	9.8196e+1 (3.76e+1)
CMIES16	3.9016e+2 (5.38e+1) -	6.4194e+2 (1.26e+2) -	4.6695e+2 (8.14e+1) -	3.3473e+2 (3.92e+1) -	2.8255e+2 (4.50e+1) -	7.5494e+1 (3.13e+1) +	3.0581e+2 (1.11e+2) -	1.3772e+2 (7.96e+1)
CMIES17	5.1879e+2 (9.97e+1) -	7.0230e+2 (1.29e+2) -	5.4364e+2 (7.85e+1) -	4.2274e+2 (5.82e+1) -	3.4380e+2 (3.40e+1) -	1.5120e+2 (4.57e+1) -	2.9594e+2 (6.48e+1) -	9.5717e+1 (4.01e+1)
CMIES18	4.2133e+2 (6.16e+1) -	6.8411e+2 (1.23e+2) -	4.8298e+2 (6.85e+1) -	3.9569e+2 (5.49e+1) -	3.1011e+2 (3.96e+1) -	1.1973e+2 (3.64e+1) -	3.1379e+2 (4.65e+1) -	9.2363e+1 (5.68e+1)
CMIES19	4.5949e+2 (7.55e+1) -	6.8250e+2 (8.98e+1) -	4.8313e+2 (6.74e+1) -	4.2360e+2 (5.75e+1) -	3.2982e+2 (4.35e+1) -	1.5078e+2 (4.99e+1) -	3.0819e+2 (4.34e+1) -	9.1730e+1 (2.89e+1)
CMIES20	4.6844e+2 (9.68e+1) -	6.9928e+2 (1.14e+2) -	5.0123e+2 (6.16e+1) -	3.6699e+2 (6.11e+1) -	3.2634e+2 (3.88e+1) -	1.1320e+2 (3.56e+1) -	3.0280e+2 (6.16e+1) -	7.8747e+1 (4.09e+1)
CMIES21	4.5419e+2 (7.15e+1) -	6.9356e+2 (1.17e+2) -	4.9026e+2 (6.59e+1) -	3.6081e+2 (5.20e+1) -	3.1197e+2 (3.10e+1) -	9.5568e+1 (4.07e+1) =	3.2076e+2 (8.08e+1) -	9.7890e+1 (3.68e+1)
CMIES22	4.4490e+2 (7.10e+1) -	7.4120e+2 (1.44e+2) -	4.8054e+2 (7.29e+1) -	3.9700e+2 (6.49e+1) -	3.3934e+2 (4.18e+1) -	1.4823e+2 (4.16e+1) -	2.9730e+2 (6.16e+1) -	1.1711e+2 (5.85e+1)
CMIES23	4.3285e+2 (5.00e+1) -	7.1714e+2 (1.10e+2) -	4.6443e+2 (5.80e+1) -	3.6767e+2 (5.55e+1) -	2.9710e+2 (3.57e+1) -	1.3012e+2 (3.59e+1) -	2.9245e+2 (1.04e+2) -	1.0229e+2 (5.84e+1)
CMIES24	4.2877e+2 (6.82e+1) -	6.7982e+2 (8.76e+1) -	4.6875e+2 (5.95e+1) -	3.6054e+2 (5.95e+1) -	2.9974e+2 (3.77e+1) -	1.1073e+2 (3.17e+1) -	2.9689e+2 (7.77e+1) -	1.0326e+2 (6.89e+1)
CMIES25	4.1157e+2 (5.63e+1) -	6.8259e+2 (1.28e+2) -	4.4299e+2 (5.69e+1) -	3.6392e+2 (4.42e+1) -	3.2943e+2 (4.30e+1) -	1.1863e+2 (3.62e+1) -	3.2023e+2 (5.41e+1) -	1.0905e+2 (5.43e+1)
CMIES26	4.0675e+2 (6.93e+1) -	6.9247e+2 (1.29e+2) -	4.4486e+2 (5.22e+1) -	3.3458e+2 (6.59e+1) -	2.9078e+2 (4.91e+1) -	1.0713e+2 (3.36e+1) -	2.9531e+2 (7.86e+1) -	8.2849e+1 (4.56e+1)
CMIES27	4.3690e+2 (6.24e+1) -	7.1931e+2 (1.19e+2) -	4.6450e+2 (6.53e+1) -	3.9755e+2 (6.37e+1) -	3.1259e+2 (4.23e+1) -	1.2121e+2 (3.08e+1) -	3.1601e+2 (4.93e+1) -	1.0211e+2 (5.23e+1)
CMIES28	3.9901e+2 (6.04e+1) -	6.5595e+2 (1.33e+2) -	4.6346e+2 (4.76e+1) -	3.7633e+2 (5.46e+1) -	3.0962e+2 (4.89e+1) -	1.1268e+2 (3.06e+1) -	3.0002e+2 (5.92e+1) -	9.3485e+1 (5.89e+1)
CMIES29	4.3251e+2 (7.06e+1) -	6.6685e+2 (1.13e+2) -	5.2066e+2 (6.11e+1) -	4.0831e+2 (7.11e+1) -	3.3309e+2 (3.74e+1) -	1.2060e+2 (4.24e+1) -	2.9495e+2 (6.64e+1) -	8.2535e+1 (3.26e+1)
CMIES30	4.4489e+2 (8.36e+1) -	6.4816e+2 (1.01e+2) -	4.8790e+2 (6.94e+1) -	3.9797e+2 (5.88e+1) -	3.2646e+2 (4.99e+1) -	1.4117e+2 (2.87e+1) =	3.2958e+2 (8.69e+1) -	1.2418e+2 (5.96e+1)
+/-/=	0/30/0	0/30/0	0/30/0	0/30/0	0/30/0	2/19/9	0/30/0	

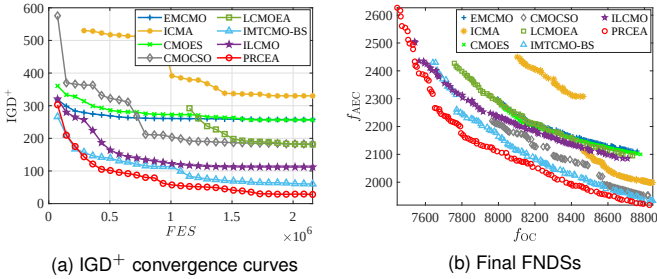


Fig. 6. IGD⁺ convergence curves and final FNDSSs obtained by eight algorithms on CMIES1 with the median IGD⁺ value among 31 runs.

for energy consumption analysis. The dispatch results and analysis of these two solutions are provided in Section S-VI of the supplementary file.

VII. CONCLUSIONS AND FUTURE WORK

This paper develops a promising region-guided algorithm named PRCEA for LSCMOPs. To prevent the population from getting trapped in local regions and obtain the complete CPF, a new PRDD strategy is introduced in PRCEA to guide the population to focus on searching the promising region that is mutually non-dominated with all well-converged FNDSSs. Specifically, PRDD integrates three dominance relations: EPD, ERPD, and PD, where EPD promotes solution diversity within the promising region, while ERPD and PD provide driving forces toward the promising region from various search directions. Moreover, an AES strategy is proposed to improve reproduction quality and accelerate convergence in large-scale decision spaces. Compared with seven state-of-the-art algorithms, PRCEA demonstrates superiority in solving most of the adopted test instances (with dimensions ranging from

100 to 1000) and 30 large-scale CMIES dispatch optimization problems.

Despite the superior performance of PRCEA, further improvements can be made in the following areas:

- 1) When solving problems with discrete CPFs, more solutions may perform inefficient searches in regions without CPF fragments, incurring unnecessary computational costs. To optimize resource allocation, a more efficient method to accurately identify regions containing CPF fragments warrants further investigation.
- 2) PRCEA struggles with some complex problems and yields poor metric values. Therefore, it is imperative to conduct further analysis of the challenges posed by LSCMOPs and research on more effective mechanisms.

The MATLAB source code of PRCEA and the raw experimental results are publicly available at: <https://github.com/rotarygirl-zxy/PRCEA>.

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